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Beijing's "14th Five-Year Plan" Plan for Climate Change and Energy Conservation

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To the people's governments of all districts, relevant municipal departments and related units:

With the approval of the municipal government, the Beijing Municipal Government's "14th Five-Year Plan for Climate Change and Energy Conservation" is now issued to you. Please implement it conscientiously.

Please be notified.

Beijing

Beijing Development and

ÿÿ2022

Beijing's "14th Five-Year Plan" Plan for Climate Change and Energy Conservation

Beijing Municipal Bureau of Ecology and Environment

Beijing Municipal Development and Reform Commission

July 2022

Preface

"China will strive to achieve carbon peak before 2030 and carbon neutrality before 2060," General Secretary Xi Jinping's solemn oath demonstrates China's positive

The firm determination to tackle climate change, strive to build a community with a shared future for mankind, and take the path of green and low-carbon development will lay a solid foundation for my country to cope with climate change in the future.

As the capital city, Beijing should take carbon peak and carbon neutrality as an important goal for planning the future development of the capital city and showcasing the city's international image.

We should fully grasp the opportunities of the times, coordinate the inherent connection between economic and social development, energy conservation and carbon reduction, and actively respond to climate change, and deepen the

Revolution, promote high-quality economic development and high-level ecological environment protection. In accordance with the overall deployment of the country and the city to respond to climate change and energy conservation,

After reaching the peak, carbon emissions have been steadily decreasing. In the process of carbon peak and carbon neutrality practice, Beijing's experience has been explored and Beijing's strength has been contributed to ensure the response to climate change.

In order to continue to be at the forefront of the country in our work, this plan has been specially formulated.

This plan is mainly based on the Beijing Urban Master Plan (2016-2035) and the 14th Five-Year Plan for National Economic and Social Development of Beijing.

Outline of the 2035 Vision Goals, Beijing Ecological and Environmental Protection Plan for the 14th Five-Year Plan Period, and Beijing Energy Development Plan for the 14th Five-Year Plan Period.

system.

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Chapter 1 Seizing New Opportunities and Starting a New Journey

1. Development Review

During the 13th Five-Year Plan period, the city has made in-depth efforts to address climate change and energy conservation, taking the lead in establishing a "dual control" mechanism for energy consumption and carbon emissions.

Promote energy conservation, greenhouse gas emission control and pollution prevention, and comprehensively apply market mechanisms, standard guidance, technology promotion and engineering support

The main targets of the plan have been exceeded, carbon emissions from energy activities have entered a peak plateau period, and the energy sector will reach a peak in 2020.

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The energy consumption per unit of regional GDP and carbon dioxide emissions were reduced by more than 23% and 26% respectively.

The carbon emission intensity remains at the best level among provincial regions in the country.

1. A green, low-carbon, high-quality development pattern has basically taken shape

We have thoroughly implemented the city's strategic positioning, taken the lead in establishing the concept of reducing development, and taken solid steps towards high-quality economic development.

Restriction list, and dynamically adjust. Strictly carry out energy-saving review of fixed asset investment projects, prohibit the construction of high energy consumption and high emission projects from the source.

2,154 general manufacturing and polluting enterprises that do not meet the functions of a capital city were removed, and industrial energy consumption and carbon emissions continued to decline. Actively develop green manufacturing systems

67 national green factories have been established. The advantages of modern service industries such as finance, information, and technology have become more prominent, and the added value of the digital economy accounts for 2.5% of the region's total output.

The city's GDP has surpassed two trillion yuan levels, and its per capita GDP and total labor productivity remain at the forefront of the country.

(II) Significant progress has been made in the process of clean and low-carbon energy

The coal consumption has been greatly reduced, and the plain areas have basically achieved coal-free. The total coal consumption in the city has been reduced from 11.65 million tons in 2015 to 10.65 million tons in 2020.

tons, and the proportion of coal in total energy consumption fell to 1.5%. Natural gas consumption accounted for 37.2%, and electricity consumption accounted for 27.8%.

The structural optimization and transformation has basically completed the clean adjustment of fossil energy. We will develop photovoltaic, wind power, geothermal and other renewable energy according to local conditions and actively

Green electricity, formulate and implement strong support policies for photovoltaic and wind power, and the proportion of renewable energy reaches 10.4%.

(III) Energy conservation and carbon reduction effects in key areas are fully demonstrated

Actively promote green buildings, build a total of 128 million square meters of green buildings, and promote ultra-low energy consumption buildings of 530,000 square meters, steadily advancing

The construction industry has been developing rapidly, with the total area of newly built prefabricated buildings exceeding 54 million square meters. Low-carbon transportation has been vigorously developed, with the operating mileage of rail transit reaching 726 kilometers.

The number of vehicles reached more than 400,000. We built a market-oriented green technology innovation system, promoted more than 200 advanced and applicable energy-saving technology products, and implemented a new

The consumption promotion policy will continue to increase the market share of energy-saving home appliances. A new round of afforestation and greening tasks of one million mu and forest carbon sink enhancement projects will be completed.

To 44.6%, forest carbon sink storage increased by about 2 million tons.

(IV) Continuous improvement of energy conservation and carbon reduction management measures

The Municipal Committee of Ecology and Civilization Construction Committee was established, with a comprehensive air pollution control and climate change response working group to promote green development.

The working group on the implementation of the project has established a working mechanism for cooperation among various districts and departments.

The industry competent departments will conduct annual assessments. We have strengthened the regulatory and binding role of standards and issued nearly 100 regulations in the fields of construction, transportation, industry, etc.

Carbon local standards. Support the promotion of contract energy management model, and continue to carry out energy audits, clean production audits, etc. Strengthen energy and carbon emissions

Capacity building has been carried out, and a carbon emission accounting system has been initially established at the city and district levels.

5. The carbon emissions trading system is becoming more mature

The implementation of the carbon emission trading system under the total carbon emission control has formed a system based on local laws and government regulations, with multiple standards and regulations.

Carbon trading policy and regulatory system. By setting emission control targets for key emission units and tightening them year by year, it can stimulate the emission units' voluntary emission reduction motivation and promote

By the end of the 13th Five-Year Plan, the city's pilot carbon market had completed seven compliance cycles, with a cumulative trading volume exceeding 35 million tons.

The total carbon emissions of enterprises included in the carbon market have accumulated over the past five years.

The carbon emission rights trading has become an important market-based means for the city to achieve green and low-carbon development, and also paved the way for the smooth launch of the national carbon market.

Beijing experience.

(VI) A good atmosphere of public participation has gradually taken shape

Organize a series of publicity activities such as Energy Saving Week, Low Carbon Day, and Low Carbon Environmental Protection Competition to actively promote the joint governance of the whole people and promote green living and green consumption

The carbon-free universal mechanism encourages citizens to fully participate in green and low-carbon travel. We have carried out extensive international exchanges and successfully held the Second China-US Climate Smart/Low Carbon

The Beijing International Metropolitan Clean Air and Climate Action Forum was held, and the United Nations Environment Programme, the United Nations Climate Conference and

We will also participate in relevant activities of international non-governmental organizations such as the Alliance for Future Leaders to tell the story of Beijing's green and low-carbon development. We will also carry out energy conservation and resource conservation activities for primary and secondary school students.

We will organize educational practice activities to enhance awareness of energy conservation and carbon reduction and social responsibility. We will issue energy conservation and carbon reduction campaigns to encourage the public to practice a green lifestyle.

The achievements made by the city in addressing climate change and energy conservation during the 13th Five-Year Plan period have laid a solid foundation for promoting green and low-carbon development during the 14th Five-Year Plan period.

There is still a huge gap between the requirements of the current carbon neutrality vision and the goal of building an international first-class harmonious and livable city.

There is still a certain gap between the indicators of urban life, energy efficiency, utilization rate of new energy and per capita carbon emissions and those of international metropolises. Secondly, the policy system is not perfect and the procedures are incomplete.

Insufficient, lack of key low-carbon technologies that can be promoted on a large scale, insufficient work team, capacity, and financial support, informatization based on social big data

The carbon emission monitoring and management system has not yet been established; third, the coordination of climate change response, energy conservation and consumption reduction, and pollution prevention needs to be further improved;

The ability to adapt to climate change needs to be continuously improved, and the ability to respond to extreme climate events needs to be continuously strengthened.

2. The current situation

1. Addressing climate change is facing increasingly severe and urgent situations and pressures

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Climate change is the most far-reaching global environmental issue. If effective policies are not taken immediately, the global average surface temperature will rise by 1.5% by 2100.

The temperature will rise by 3.7-4.8ÿ, causing irreversible global disasters. At present, the efforts of various countries are not enough to support the Paris Agreement on Climate Change.

The goal is to control global temperature rise at 2°C by the end of this century and strive to keep it at 1.5°C. As a country that is significantly affected by climate change, China has recently

The trend is obvious, and various climate disasters have brought huge losses. As a megacity, the average temperature of our city has been

0.51ÿ, which is higher than the national warming rate during the same period. Climate warming and urban heat island effect have led to frequent extreme climate events in the city, with extreme

The extreme disasters, health threats, and resource depletion caused by climate change are of great significance to the sustainable development of cities,

The contradiction between the desire for a good life and the desire for a healthy life is becoming increasingly prominent, and the task of responding to climate change in the capital is extremely arduous.

(II) New requirements for addressing climate change and energy conservation

Striving to achieve carbon peak by 2030 and carbon neutrality by 2060 is a major strategic decision made by the CPC Central Committee after careful consideration.

Sustainable development and building a community with a shared future for mankind. The proposal of the "dual carbon" goal has brought new opportunities and historical missions to the development of the capital.

We have entered a critical window period for implementing major decisions and deployments on carbon peak and carbon neutrality. Green, low-carbon and circular development has become the main development mode for economic and social development.

Efficient utilization has become the key, and the importance of addressing climate change and energy conservation has become more prominent. In order to play the role of a model and leader in the capital, the Beijing Municipal

The Outline of the 14th Five-Year Plan for National Development and Social Development and the Long-Term Objectives for 2035 clearly states that during the 14th Five-Year Plan period, carbon emissions will peak and then stabilize and decline, and carbon emissions will

The goal of solid progress has put forward higher requirements for addressing climate change and energy conservation. We must truly take energy conservation and climate change mitigation as the main

The city will take climate change adaptation as an important part of testing the city's planning and construction and environmental governance capabilities, and make energy conservation and carbon reduction a comprehensive

Development pattern.

(III) Carbon peak and carbon neutrality goals bring new opportunities for high-quality development

my country and major international economies have successively proposed carbon neutrality goals, the essence of which is to respond to the global climate crisis through green and low-carbon transformation and development.

Development will shift to technology-dependent development, rapidly promote a new round of global green technological revolution and industrial revolution, and promote the world economy to achieve "green recovery" after the epidemic.

The capital should fully grasp the development opportunities brought by carbon peak and carbon neutrality, and unswervingly follow the high-quality development path of ecological priority, green and low-carbon.

We will leverage our scientific and technological talent advantages to quickly occupy the commanding heights of low-carbon technology and strive to cultivate new energy technology integration, new energy vehicles, fuel cells, hydrogen energy, etc.

points, promote the transformation of new and old kinetic energy, and achieve high-quality ecological environment protection and high-quality sustainable development of social economy with green and low-carbon as the main characteristics

(IV) New challenges for efficient energy use and low-carbon transformation

During the 14th Five-Year Plan period, the city will focus on building a world-class city cluster with the capital as its core, and creating a modern economic system with both characteristics and vitality.

The well-being of people has been significantly improved, the development of key industries such as air transportation, 5G communications, digital economy, and biomedicine has been accelerated, and energy consumption and carbon emissions have also been reduced.

With the gradual recovery of production and life after the epidemic, the city's total energy consumption and carbon emissions in the first two years of the "Fourteenth Five-Year Plan" will be higher than that in 202

Therefore, to achieve the energy and carbon emission control goals, it is necessary to overcome the poor endowment of new energy resources, insufficient technological maturity,

We will continue to optimize the power structure, heating mode, travel mode, and vehicle structure, gradually increase the proportion of renewable energy, and significantly improve

efficiency, plan and deploy a green electricity supply system in a larger area, accelerate the formation of a policy environment conducive to promoting carbon peak and carbon neutrality, and continue to develop

Through the development and application of technologies, we will promote the green and low-carbon transformation of the energy system with electrification and decarbonization as its main features as soon as possible on the basis of clean energy.

Chapter 2 Seizing New Opportunities and Starting a New Journey

1. Guiding Ideology

We will continue to be guided by Xi Jinping's Thought on Socialism with Chinese Characteristics for a New Era, fully implement the spirit of the 19th CPC National Congress and all previous plenary sessions of the 19th CPC Central Committee, and deepen

Xi Jinping's Thought on Ecological Civilization and the spirit of General Secretary Xi Jinping's series of important speeches on Beijing, fully and accurately implement the new development concept, and plan around

The capital's economic and social development should be properly coordinated with the inherent connection between resource and energy utilization, environmental quality improvement and climate change response, so as to actively respond to the

The goal is to take the lead in achieving a steady decline in carbon emissions after reaching the peak, promote the overall green transformation of the economy and society, and focus on low-energy

As the key, we will improve the total amount and intensity control mechanism of energy and carbon emissions, coordinate the use of legal, standard, market and technical means, and strive to enhance

Energy efficiency, actively control greenhouse gas emissions, and continuously improve the city's ability to adapt to climate change, in order to achieve high-quality ecological environment in the capital.

Lay a solid foundation for high-quality social development, fulfill the country's commitment to independent contribution, actively participate in global climate governance, and strive to achieve the goal of controlling global temperature rise.

Beijing contribution.

2. Basic Principles

Overall planning and coordinated advancement. We should properly handle the relationship between development and emission reduction, adhere to resource conservation, high-quality ecological environment protection, and actively respond to climate change.

High-quality economic and social development is coordinated and consistent. Vigorously promote breakthroughs in energy-saving and low-carbon technology innovation, and form an endogenous dynamic of low-carbon leading energy and industrial revolution.

Green, low-carbon, circular development economic system. Coordinated control of greenhouse gas and pollutant emissions to achieve synergistic effects of pollution reduction and carbon reduction.

Prioritize conservation and make breakthroughs in key areas. Handle the relationship between the whole and the local areas, adhere to the principle of "one country, one chess game", and give full consideration to the city's industrial structure and energy consumption.

We will focus on key areas, highlight the key role of energy conservation, strictly control fossil energy consumption, and promote the greening of industries as a means to achieve economic development.

It will improve efficiency, promote urban renewal with energy-saving and low-carbon transformation of buildings, and promote sustainable development of transportation with the focus on electrification of motor vehicles.

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We should focus on the present and the future. We should handle the relationship between the short-term and the medium- and long-term well, insist on linking the five-year plan with the long-term development strategy, and grasp the Rhythm and strength. Anchored on the long-term carbon neutrality goal, based on the current work reality, pragmatic, step-by-step, and continuous efforts, accelerate the implementation of energy conservation We will use such no-regrets measures to implement the energy conservation and carbon reduction targets during the 14th Five-Year Plan period and ensure that carbon emissions continue to decline steadily after reaching their peak.

Two-wheel drive, systematic policy implementation. Handle the relationship between the government and the market well, and insist on exerting force on both sides. Give full play to the guiding role of laws and regulations and strengthen Improve the total amount and intensity control mechanism of energy and carbon emissions. Strengthen the main responsibility of emission units, improve the carbon emission rights trading mechanism, and form an effective Mechanism. Create a policy environment conducive to promoting energy conservation and carbon reduction, and advocate a simple, moderate, green and low-carbon social trend.

3. Main Objectives

By 2025, we will actively promote the transition from dual control of energy consumption to dual control of total carbon emissions and intensity, and form a current situation that is basically adapted to the characteristics of megacities. With the climate change governance system and energy-saving policy and measure system, cities' ability to adapt to climate change has been significantly improved. Energy efficiency and carbon emission level protection in key areas First, carbon neutrality has taken a solid step forward, laying the foundation for actively responding to the challenges of climate change and achieving the carbon neutrality goal.

——The total carbon emissions peaked first and then declined steadily. It dropped by more than 10% from the peak (excluding carbon emissions from air passenger and cargo transportation), per unit area The reduction rate of carbon dioxide emissions has reached the national requirement. Non-carbon dioxide greenhouse gas emissions have been actively controlled.

——The efficiency of energy resource utilization has been greatly improved. The total energy consumption of the city is controlled at about 80.5 million tons of standard coal, and the gross regional product per unit area is 14%, and the new energy consumption will be guaranteed by renewable energy first.

——The proportion of low-carbon energy has been greatly increased. Coal is basically not used in non-emergency situations, and the total coal consumption is controlled within 1 million tons. The proportion of renewable energy will be increased to more than 14.4%, and the installed capacity of local renewable energy will reach 4.35 million kilowatts. Strive to reach 30 billion kWh.

——Continuously improve the carbon sink capacity of the ecosystem. The carbon sink capacity has been greatly improved, the forest coverage rate has increased to 45%, and the forest stock volume has increased to 30%. rice.

——The ability to adapt to climate change has been greatly improved. A strong and resilient urban lifeline has been built, and the city's ability to cope with extreme climate events has continued to improve.

Vision goals for 2035: The total carbon emissions will drop significantly, the total energy consumption will enter the peak plateau period, and the energy resource utilization efficiency will reach the international level, the proportion of renewable energy has increased significantly. The quality of the ecological environment has fundamentally improved, and significant results have been achieved in combating climate change.

Table 1 Indicators for climate change response and energy conservation planning during the 14th Five-Year Plan period

Serial number	index	Planning value in 2025	refer to
1	Total carbon emissions (billion tons)	Compared with the peak (excluding aviation), it has dropped by more than 10%.	
2. Reduction in carbon dioxide emissions per unit of GDP (%)		Meet national requirements	about
3	Total energy consumption (10,000 tons of standard coal)	Around 8050	Pre
4	Reduction in energy consumption per unit of regional GDP (%)	14%	about
5	Renewable energy consumption ratio (%)	14.4% or more	about
6	Renewable energy heating area share (%)	10	Pre
7	Coal consumption (10,000 tons)	<100	about
8	Natural gas consumption (billion cubic meters)	Around 200	Pre
9	Green electricity transferred externally (billion kWh)	Strive for 300	Pre
10	Local renewable energy installed capacity (10,000 kW)	435	Pre
11	Forest cover rate(%)	45	Pre
12	Forest volume (10,000 cubic meters)	3000	Pre
13 Decrease in carbon dioxide emissions per unit of industrial added value (%)		20	Pre
14	Decrease in energy consumption per unit of industrial added value (%)	12	about
15	Ultra-low energy consumption building scale (10,000 square meters)	500	Pre
16	Area of public building energy-saving renovation (10,000 square meters)	3000	Pre
17	Total energy consumption of public institutions (10,000 tons of standard coal)	<165 (equivalent value)	Pre

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18	Total carbon emissions from public institutions (10,000 tons)	<571	Pre
19	Green travel ratio in central urban areas (%)	76.5	about
20	Proportion of area meeting sponge city standards (%)	ý40	Pre

Chapter 3 Comprehensively promote green, low-carbon and circular development

We will take climate change response and energy conservation as an important way to promote high-quality development, fully implement the strategic positioning of the capital city, and thoroughly implement the reduction of

We will promote the green and low-carbon transformation of industrial structure, energy structure and urban functions, accelerate the release of new momentum for green development, and vigorously promote

Regional green development.

1. Deeply promote the optimization and adjustment of industrial structure

1. Systematically promote the upgrading of industrial structure

Actively develop high-tech, low-resource consumption, and low-carbon industries, and accelerate the transformation of high-energy-consuming and high-emission manufacturing enterprises and production

The government will eliminate high-energy-consuming equipment and orderly guide the transfer and withdrawal of general manufacturing industries that do not conform to the functional positioning of the capital, strictly control and reduce petrochemical production in Beijing,

Revise and strictly implement the list of prohibited and restricted industries, design entry conditions in accordance with advanced energy-saving and low-carbon standards, and resolutely control high-energy-consuming

Strengthen the coordinated control of environmental impact assessment, energy assessment and carbon assessment at the source, and strictly control the energy consumption and carbon emission levels of new projects.

(II) Deeply stimulate new momentum for green development

Lead the development of high-tech industries with green technology innovation, continue to promote the construction of a green manufacturing system, and strive to cultivate new energy, new materials, and new energy.

Add new strengths to fuel cells, hydrogen energy and other industries. Build new energy industry clusters and industrialization demonstration bases to enhance the modernization of industrial chains and supply chains.

Actively create a national green industry base and build an internationally competitive green industry cluster. Cultivate and develop new formats of green smart technology services.

By 2025, a modern green economic system with capital characteristics will be basically formed.

(III) Continue to promote green transformation of industries

Benchmarking against international and domestic advanced energy efficiency and carbon emission levels, we will implement a number of energy-saving and low-carbon technology transformation projects. We will carry out green transformation of key industries.

Recycle waste heat and pressure from power plants, gas boilers, industrial kilns, sewage treatment plants, data centers, etc., and accelerate the implementation of energy-saving transformation of road lighting.

We will continue to implement energy efficiency standards for data centers and continue to carry out energy-saving and carbon-reduction transformation of data centers. We will vigorously develop a circular economy, comprehensively promote clean production, and promote resource

Carry out green and low-carbon recycling transformation of parks at or above the municipal level, and strengthen the cascade utilization and system optimization of park resources. Carbon emissions from manufacturing industry during the "14th Five-Year Plan" period

The increase and decrease were balanced, with energy consumption per unit of industrial value added and carbon dioxide emissions falling by more than 12% and around 20% respectively.

Column 1 Energy-saving and low-carbon transformation of data centers

In order to promote the continuous improvement of energy efficiency and carbon efficiency of data centers from the source, strengthen life cycle energy conservation management, and promote high-quality development of data centers

To support the construction of a digital economy benchmark city, the city has formulated and issued the "Several Provisions on Further Strengthening the Energy-Saving Review of Data Center Projects"

The energy efficiency level (PUE value) standard for new, renovated and expanded data centers has been determined. The closer the value is to 1, the higher the energy efficiency level.

The PUE value of projects with an annual energy consumption of 10,000 tons of standard coal is not higher than 1.3; the PUE value of projects with an annual energy consumption of 10,000 tons of standard coal and less than 20,000 tons of standard coal is

The PUE value of projects with annual energy consumption greater than or equal to 20,000 tons of standard coal and less than 30,000 tons of standard coal shall not be higher than 1.2; the PUE value of projects with annual energy consumption greater than or equal to 20,000 tons of standard coal shall not be higher than 1.2.

The PUE value of projects with an energy consumption of 30,000 tons of standard coal shall not be higher than 1.15.

Data centers should make full use of natural cooling sources and strengthen the use of waste heat resources through self-use and external heating.

The energy management system should be established and improved in accordance with the relevant standards such as "Energy Management System" (GB/T23332). Actual operation of the data center

Data centers that exceed the standard energy efficiency limit will be charged differential electricity prices.

The city's pilot carbon market focuses on strengthening carbon emission management in the data center of key emission units, formulating and dynamically updating the city's data center carbon emission management system.

Value and advanced value, promote data centers to reduce carbon emissions in a market-oriented way, energy efficiency levels are poor, carbon emissions exceed the quota issued by the competent authorities

The central operating unit needs to purchase quotas from the market to fulfill its carbon emission control responsibilities.

2. Deepen the low-carbon transformation of energy structure

1. Control the scale of fossil energy utilization

We will vigorously promote the use of electricity to replace coal, orderly promote the use of electricity to replace oil and gas, strictly control the growth of fossil energy consumption, and ensure that the total carbon emissions of energy activities are

Accelerate the reduction of industrial coal use, use existing coal-fired units as emergency backup units, and orderly promote the replacement of coal with clean energy in mountainous villages.

Realize "coal-free" villages in shallow mountainous areas, basically no longer use coal in non-emergency situations, and control the total coal consumption within 1 million tons. Promote oil products

Steady but declining, accelerate the replacement of motor vehicles with electric power, and control the rapid growth of aviation kerosene. Control the scale of natural gas utilization and the total amount of natural gas consumption.

About square meters.

(II) Vigorously promote the use of renewable energy

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Develop the city's renewable energy according to local conditions, vigorously develop photovoltaics and geothermal energy, moderately develop wind power and waste incineration power generation, and promote hydrogen energy demonstration

Promote the use of distributed power generation on-site and grid-connected, and actively develop "new energy + energy storage", "source-grid-load-storage" and multi-energy complementarity.

First, use renewable energy to ensure supply, improve the infrastructure and industrial support for renewable energy utilization, and promote the integration of renewable energy, new technologies and cities.

Development. By 2025, the proportion of renewable energy in the city will reach more than 14.4%. Vigorously develop clean heating models such as geothermal and heat pumps, solar energy, and energy storage and heat storage.

The proportion of heating area from renewable energy in the city reached more than 10%.

3. Promoting low-carbon energy transformation in the Beijing-Tianjin-Hebei region

Strengthen the integrated development of energy in Beijing, Tianjin and Hebei, and strengthen cooperation in low-carbon energy development. Build cross-regional power, natural gas, oil and other energy transmission

Promote the interconnection of energy infrastructure in the region. Jointly develop wind and solar energy resources in the region, giving priority to renewable energy and clean energy.

Source on the Internet. Build Zhangjiakou Renewable Energy Demonstration Zone with high standards. Promote the cooperation and construction of new energy bases between Beijing and Hebei, and accelerate the construction of new energy bases in the northwest and south.

We will build power transmission channels and distribution peak-shaving energy storage facilities to effectively improve the city's green power consumption capacity. By 2025, the city's external green power will reach

Orderly layout of hydrogen energy industry development, with scientific and technological innovation as the core, strengthen policy guidance and industry cultivation, and drive the coordinated development of hydrogen energy industry in Beijing, Tianjin and Hebei

We strive to build a complete hydrogen energy and fuel cell industry chain.

III. Solidly promote the transformation and upgrading of urban functions

1. Leading the green upgrade of urban planning and construction

Continue to optimize the layout of urban functions, unswervingly relieve non-capital functions, build a system and mechanism to promote reduced-volume development, and reasonably control the development of strong

Incorporate the concept of green and low carbon into the overall layout of urban planning and construction. Take urban renewal as an opportunity to promote the green upgrade of urban construction.

Taking into account the requirements of green, low-carbon, circular development and adaptation to climate change, we will promote the upgrading and transformation of infrastructure such as transportation, energy, and water treatment, and strengthen the construction of new energy.

Use it to build strong, resilient, low-carbon and smart urban infrastructure.

(II) Promoting differentiated development in key regions

Based on the functional positioning of different regions, taking into account factors such as economic and social development level, resource endowment, energy-saving potential, etc., regional differentiation is implemented.

The central urban area should continue to relieve non-capital functions, promote urban renewal with energy conservation and carbon reduction as the orientation, strictly control the growth of total energy consumption, and reduce carbon emissions.

Pingyuan New City should strengthen the application of technology demonstration, implement energy consumption and carbon emission control mechanisms, focus on improving energy efficiency, and take the lead in

Development transformation. Ecological conservation areas should focus on the large-scale development and application of renewable energy, and strive to achieve a stable and declining carbon emissions platform period.

(III) Comprehensively promote the transformation and optimization of regional functions

Give full play to the radiation and driving role of Beijing as the "core", accelerate the construction of a modern capital metropolitan area, and promote the formation of the main framework of the Beijing-Tianjin-Hebei urban agglomeration.

Function, promote the linkage between Beijing's sub-city center and Hebei Xiong'an New Area. Promote the planning and construction of the sub-city center at a high level around the green and low-carbon concept.

We will build green development demonstration areas, near-zero carbon emission demonstration parks and characteristic towns. We will optimize the regional pattern and industrial policies for low-carbon development, and do a good job in regional

Transfer, undertake, transform and upgrade, and promote the formation of an industrial system characterized by green and low-carbon. Cooperate to expand green ecological space and actively develop regional forests.

Aim: To promote the organic combination of carbon trading mechanism and ecological compensation.

Chapter 4 Strictly Control Greenhouse Gas Emissions in Key Areas

Focusing on key areas such as construction and transportation, we will accelerate the application of advanced energy-saving and low-carbon technologies, implement energy-saving and carbon-reduction projects, and highlight refinement and intelligence.

Effectively improve energy efficiency levels and strictly control emissions of various greenhouse gases.

1. Comprehensively promote low-carbon development in the construction sector

1. Strictly implement energy-saving standards for new buildings

Residential buildings will be the first to implement the 80% energy-saving design standard. All new civil buildings will implement green building standards. Accelerate the revision of public building energy-saving

New government-invested public welfare buildings and large public buildings must meet the green building two-star standard or above.

For energy building applications, new buildings that meet the conditions should be equipped with photovoltaic or solar thermal systems.

(II) Vigorously promote ultra-low energy consumption buildings

Establish and improve the city's ultra-low energy consumption building policy and standard system, formulate ultra-low energy consumption public building design, construction, and acceptance standards, and

We will vigorously promote ultra-low energy consumption buildings in public building construction. We will encourage the promotion of ultra-low energy consumption buildings in commercial residential construction, and in sub-centers of cities, "three cities" and "three

Promote ultra-low energy consumption buildings in key areas and functional parks such as the air-space economic zone. Vigorously develop prefabricated buildings and strengthen systematic integrated design.

The cumulative promotion scale of energy-consuming buildings has reached more than 5 million square meters.

Column 2 Ultra-low energy consumption buildings

The construction sector is an important part of energy consumption and carbon emissions in human activities. Continuously promoting building energy conservation and reducing building carbon emissions is the future.

An important measure for sustainable development. At present, countries around the world are vigorously promoting the process of buildings moving towards near-zero energy consumption. Ultra-low energy buildings refer to buildings that

Climate characteristics and site conditions, make full use of natural ventilation, natural lighting, thermal insulation of enclosure structures and other technical measures, adopt high-efficiency energy

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Buildings that minimize energy consumption for heating, air conditioning, lighting, etc. On the basis of ultra-low energy buildings, by adding renewable energy buildings

Technical measures can achieve near-zero energy operation and minimize building carbon emissions.

The development goal of ultra-low energy consumption buildings in my country is to build 30% of new buildings into ultra-low energy consumption buildings in the future, and renewable energy should meet the needs of new buildings.

30% of energy consumption, 30% of existing buildings should be renovated to achieve ultra-low energy consumption levels. Since the 1980s, Beijing has successively issued

30%, 50%, 65%, 75% design standards, and in 2021, the city will be the first in the country to implement the fifth part of the design standard for residential buildings to achieve 80% energy efficiency.

During the five-year plan, the city will promote ultra-low energy consumption buildings covering an area of more than 5 million square meters, complete energy-saving renovation of 30 million square meters of public buildings, and implement construction

The installed capacity is 800,000 kilowatts, and we strive to achieve a balance between increases and decreases and a steady decline in carbon emissions in the construction sector by the end of the 14th Five-Year Plan.

3. Promote energy-saving renovation of existing buildings

Establish a long-term working mechanism for greening existing buildings, conduct surveys on existing non-energy-efficient buildings, focus on public buildings, and increase greening efforts in existing buildings.

We will strengthen the transformation of urban areas and complete the energy-saving transformation of 30 million square meters of public buildings. We will further strengthen the comprehensive renovation and upgrading of old urban areas.

Complete the energy-saving renovation of existing non-energy-saving residential buildings and rural houses in cities and towns, and strive to basically complete the renovation of old and dilapidated small buildings in cities and towns built before 2000.

Promote energy-saving transformation of heating systems, optimize the layout of heating network facilities in central urban areas, make full use of waste heat resources, and comprehensively deploy new energy heating.

We will carry out infrastructure renovation and upgrading projects and clean heating, further promote the greening of rural residential and public buildings, and improve the energy efficiency of heating.

(IV) Strengthen the refined management of building operation energy consumption

Strengthen the comprehensive management of building energy consumption and carbon emissions, and establish a data system covering all aspects from energy supply to energy consumption. Strengthen energy conservation in public buildings

Improve the clean heating system and promote the use of heating boilers and

Intelligent operation and management, all new buildings will have heat metering systems, and public buildings with centralized heating will implement a heat metering and charging system.

The heating energy consumption of building area will drop by about 10%, and the total carbon emissions in the construction sector will strive to achieve a balance between increase and decrease, with a steady decline.

2. Strive to build a low-carbon transportation system

1. Actively guide green travel

Build slow-travel-friendly cities, promote the integrated development of various modes of transportation, and build a green travel system that is convenient, interconnected, and multi-network integrated.

The proportion of green travel in urban areas reached 76.5%. Priority will be given to the development of urban public transportation, improving the connection system between rail transportation and ground bus systems, and promoting

Through energy-saving technology and equipment, a number of energy-saving and low-carbon rail transit demonstration lines will be built.

(II) Accelerate the optimization of vehicle structure

Formulate and implement the work plan for promoting the application of new energy vehicles during the 14th Five-Year Plan period, improve the total motor vehicle control and traffic management policies, and accelerate the promotion of new energy vehicles.

The city will strive to promote the use of new energy vehicles to replace fuel vehicles, and strive to reach 2 million new energy vehicles by 2025.

Public transportation, taxis and other public vehicles have basically been electrified. In accordance with the principle of moderate advancement, supporting facilities such as charging piles (stations), hydrogen refueling stations, etc. will be built.

Set up charging networks on urban highways. By the end of the 14th Five-Year Plan, we will strive to achieve a steady decline in carbon emissions in the transportation sector.

3. Strengthen energy conservation and carbon reduction in the aviation and cargo sectors

In the fields of aviation and railway transportation, we actively promote the whole process management of energy conservation and carbon reduction, strengthen the energy consumption and carbon emission management of airports and stations, and strengthen the new

Utilize, promote energy-saving transformation in the aviation field, promote energy efficiency improvement, and jointly reduce carbon dioxide and pollutant emissions. Improve the freight structure and enhance

Increase road transport volume and guide the effective connection between railway transport and urban distribution. Accelerate the development of intensive, low-energy green logistics. Accelerate the transformation of logistics vehicles.

Cars, widely promote urban electric logistics vehicles, and accelerate the demonstration and promotion of new energy logistics vehicles such as hydrogen energy.

3. Continue to promote energy conservation and carbon reduction in public institutions

Improve the overall coordination mechanism for energy conservation and carbon reduction in public institutions, strengthen the construction of energy conservation and resource management capabilities, and accelerate the formation of a complete energy conservation and carbon reduction system.

Implement total energy resource control, continuously improve the efficiency of energy resource utilization in public institutions, further increase the proportion of renewable energy, and widely develop

During the 14th Five-Year Plan period, the total energy consumption of public institutions was controlled within 1.65 million tons of standard coal (equivalent value).

The total emission will be controlled within 5.71 million tons. Taking 2020 as the base year, by 2025, the comprehensive energy consumption per unit building area and per capita comprehensive energy consumption will be

6%, carbon emissions per unit building area and per capita carbon emissions decreased by 9% and 7% respectively. Formulate the Energy Conservation and Low Carbon Action Plan for Party and Government Agencies, State-owned Enterprises and Institutions

The document also calls for public institutions to play a leading role in low-carbon development.

4. Controlling Non-CO_2 Greenhouse Gas Emissions

Strengthen the monitoring, statistics and scientific management of non-carbon dioxide greenhouse gases such as methane, sulfur hexafluoride, nitrous oxide and perfluorocarbons in the city.

Standards and measures for the control of non-CO_2 greenhouse gas emissions such as ethane. Promote CO_2 refrigerants. Reduce the intensity and total amount of fertilizer use, etc.

Reduce methane and nitrous oxide emissions in the field, increase the proportion of low-consumption and high-efficiency agricultural facilities, promote high-efficiency livestock and poultry breeding technologies, and improve the utilization rate of organic feed.

Box 3 Non-CO_2 Greenhouse Gas Emission Control

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Non-CO2 greenhouse gases are the other six greenhouse gases besides CO2, mainly including methane (CH4), nitrous oxide (NO (N2O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF6) and nitrogen trifluoride (NF3).

Non-carbon dioxide greenhouse gases have a strong warming effect. The fifth report of the United Nations Intergovernmental Panel on Climate Change pointed out that Since 1997, about 35% of greenhouse gas radiation has come from non-CO2 greenhouse gas emissions. Non-CO2 greenhouse gases have a long life span and warm the world. The negative impact on the earth's environment is large, and the net warming effect is about 2/3 of the warming effect of carbon dioxide, plus air pollution forms smog. The warming effect is comparable to that of carbon dioxide.

According to the National Greenhouse Gas Inventory, non-CO2 greenhouse gases account for 2.5% of my country's total greenhouse gas emissions (excluding land use change). The greenhouse gas emissions account for about 16.5% of the world's total greenhouse gas emissions, of which methane accounts for 56%, nitrous oxide accounts for 30%, and fluorinated gases account for 14%. The main emission sources are Energy activities, etc. Beijing's non-CO2 greenhouse gas emissions (excluding greenhouse gas emissions from land use change and forestry) account for a relatively low proportion. The greenhouse gas emissions account for about 4.2% of the total greenhouse gas emissions in Beijing, of which methane accounts for more than 50%, nitrous oxide and fluorinated gases account for about 20% each; for waste, industrial production processes and energy activities.

Chapter 5 Strengthening Urban Climate Adaptability

Take climate change adaptation as an important part of building resilient cities, focus on strengthening urban climate resilience, and strengthen extreme weather risk prevention The ability of high-rise cities to perceive, forecast and warn about climate will help improve the level of urban planning, construction and management.

1. Systematically improve the carbon sequestration capacity of urban ecosystems

1. Enhance the carbon sequestration capacity of forest ecosystems

Highlight the role of gardening and greening in improving urban microclimate, adhere to the multi-level greening and carbon fixation in the whole region, enhance the carbon sink capacity of forestry, and improve forest We will establish monitoring systems and assessment mechanisms, strengthen basic research capabilities on forestry carbon sinks and wetland carbon sinks, and develop ecologically friendly, high carbon sink, and highly adaptable tree species banks. Improve the efficiency and refinement of landscape management and promote the increase of carbon sinks in landscape green areas. Enhance the carbon sequestration capacity of wetlands, protect wetlands according to local conditions, and gradually Water space natural ecology. By 2025, the forest coverage rate will be increased to 45%, and the forest stock will increase to 30 million cubic meters.

(II) Strengthening the carbon sink capacity of agricultural ecosystems

Develop sightseeing agriculture, specialty agriculture, and smart agriculture, give full play to the advantages of science and technology, and promote agricultural facilities, parks, integration, greening, Establish an efficient, ecological, and energy-saving farming system, strengthen the protection of arable land resources, promote the return of straw and organic fertilizer to the fields, and increase the construction of high-standard farmland. Carry out comprehensive land management, increase soil organic carbon reserves, and enhance the carbon sequestration capacity of farmland soil.

2. Strengthening capacity building for adaptation to climate change

1. Improving the ability of urban lifelines to cope with climate change

Conduct climate change disaster risk assessments on major engineering projects in urban lifeline systems such as power supply, water supply, gas supply, and communications. Improve the construction and operation standards of urban lifeline projects, and revise the standards for the protection of urban infrastructure from meteorological disasters. The intelligent control of the gas, power and heat supply dispatching systems will be further improved, and the coordination mechanism of the dispatching system of the gas, power and heat supply will be improved, and the basic infrastructure of transportation, water supply, power supply and communication will be improved. Climate change resilience.

(II) Improving urban nature-based solutions

Build the urban green ecological background, strengthen green patches, green corridors, urban ecological safety control system, urban living environment, garden greening Airway construction. Improve the quality of attached green space and build a complete ecological network of urban forests, green space, water areas, and rivers. Focus on promoting urban ecology Climate change work, encourage and utilize nature-based solutions, and build a spatial pattern in which ecological corridors and urban construction are integrated.

3. Continue to promote adaptive urban construction

Research and construct healthy city standards for spatial planning, and assess the impact of climate change on public health, food security, urban safety, etc. Strengthen Optimize the layout and construction of "service-oriented" public service facilities. Promote the transformation of urban construction from incremental development to stock renewal, and implement urban The city has launched a sponge city construction project, including "sponge campus", "sponge park" and "sponge road". Prepare special planning for Miancheng City and implement the zoning control strategy for sponge city construction. Accelerate the implementation of various sponge city planning and construction tasks and complete Through sponge transformation, the proportion of sponge city-compliant areas in the city will reach more than 40%.

3. Establish a climate crisis prevention system

1. Conduct climate risk identification and disaster assessment

Carry out climate vulnerability risk identification and disaster assessment, improve the city-wide meteorological monitoring and natural disaster early warning system, and regularly provide water supply companies with Enterprises, communication companies and gas companies issue disastrous weather warning information. Establish a climate change data sharing platform to form cross-departmental data sharing Improve the ability of urban climate change prediction and early warning, carry out short-term climate forecasting, future scenario prediction, extreme event assessment, impact analysis and assessment, etc.

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(II) Improving early warning levels in key areas and vulnerable regions

Differentiate between conventional climate risks and extreme risks, establish a multi-sector, multi-hazard, multi-subject, and cross-regional collaborative governance and decision-making mechanism, and strive to

Improve the comprehensive climate risk warning capabilities for key areas and vulnerable areas of the city. Improve the prediction, forecast and warning capabilities of extreme climate events, and screen extreme climate events.

Region, strengthen sub-regional forecast and short-term forecast.

3. Improving emergency response capabilities for extreme climate events

Establish a climate change risk management system led by the government, coordinated by departments, and involving the public, and formulate emergency plans to resist different types of climate disasters

We will strive to build an all-weather, systematic, and modernized extreme weather risk prevention and response system.

Risk points, build a multi-level, high-resilience emergency disaster prevention space system of cities, streets and communities, and a resilient facility system that combines peacetime and wartime. Optimize the configuration of various

Improve climate disaster early warning, emergency response and disposal capabilities.

Column 4 Urban Adaptation to Climate Change

Adapting to climate change refers to the process of actively adjusting to the impacts of current and future climate change, seeking benefits and avoiding harm. The goal is to

Smart economy and building a climate-resilient society. At the city level, climate change adaptation focuses on the city system's ability to respond to future climate risks.

Successful climate change adaptation actions can effectively avoid the negative impacts of climate change and reduce the losses caused by climate risks.

Benefits from climate change.

The Beijing Urban Master Plan (2016-2035) proposes that by 2035, Beijing will basically become a world-class harmonious and livable city.

To solve the "big city disease", we should fully consider urban climate information and climate change factors in urban planning and scientifically formulate urban planning development strategies.

In the Outline of the 5th Five-Year Plan, it is clearly proposed to build resilient cities and strive to improve the ability of cities to adapt to climate change.

Encourage innovation to establish multi-goal coordinated urban management, information communication and feedback mechanisms, and intelligent management ecosystems, and adopt different policy scenarios to

The plan takes adaptation to climate change as one of the important principles for the construction of resilient cities, smart cities, sponge cities, etc.

Chapter 6: Building a comprehensive governance system for energy conservation and carbon reduction

Based on strengthening the construction of laws and regulations, focusing on consolidating the responsibilities of various entities, and improving market mechanisms such as carbon trading,

Guided by joint participation, we will accelerate the formation of a legal, market-oriented, refined and diversified urban low-carbon governance system to transform energy conservation and carbon reduction into a

The subject's conscious actions.

1. Strengthen the construction of laws, regulations, systems and standards

1. Improve laws, regulations, policies and measures

Promote Beijing's legislation on climate change response, and gradually build a legal guarantee system for climate change response oriented towards achieving carbon peak and carbon neutrality goals.

In accordance with the needs of the country and the city, the Beijing Carbon Emission Trading Management Measures (Trial) should be revised as soon as possible.

Energy Conservation Law of the People's Republic of China> Measures> Beijing Energy Conservation Supervision Measures and other relevant laws and regulations. Promote the ecological environment protection, resource and energy utilization

In the process of formulating and revising relevant laws and regulations on construction, relevant requirements for responding to climate change should be added.

Establish a supervision and law enforcement system, open up channels for public supervision, strictly supervise and enforce the law, and expose illegal activities in a timely manner.

(II) Accelerate the promotion of energy conservation and carbon reduction standardization

Improve the energy-saving and low-carbon standard system, give full play to the standard constraint and forced leading role, explore the carbon emission standard and energy efficiency standard access and industry catalog

The mechanism will focus on the carbon emission and energy consumption limit requirements of industries (products), formulate and revise a number of energy-saving and carbon-reduction standards, and gradually build a system covering all fields and all environments.

Actively improve the level of energy-saving and low-carbon development, and guide enterprises in various industries to reduce energy consumption and carbon emission intensity.

2. Strengthening target responsibility and supervision assessment

1. Strengthening the target responsibility supervision system

Further strengthen the leadership of the Municipal Committee on Ecological and Civilized Construction in the city's response to climate change and energy conservation, and coordinate all departments and units

The city's carbon emission control and energy conservation targets will be further clarified, and the division of responsibilities of key industries and districts in energy conservation and carbon reduction will be formed to promote coordinated efforts.

Implement an analysis, consultation, monitoring and early warning mechanism to timely assess the carbon emissions and energy consumption situation and conduct supervision of key regions and industries.

(II) Improve the decomposition and assessment mechanism

Strengthen the constraints of carbon emission and energy consumption control targets, and comprehensively implement the target decomposition of carbon emission total amount and intensity, energy consumption total amount and intensity

Fully consider the influencing factors such as the development stage, functional positioning, energy demand and carbon reduction potential of each district and industry, consolidate control responsibilities, and make energy conservation

The targets will be broken down vertically to each district and horizontally to the municipal industry departments, and the energy conservation and carbon reduction targets will be refined.

The nuclear system clearly states that newly added renewable energy and raw material energy will not be included in the total energy consumption control. The assessment results will be included in the district government and municipal

The performance appraisal system of the department will be improved. The appraisal method will be improved, and the transition from the "dual control" of energy consumption to the "dual control" of total carbon emissions and intensity will be actively promoted.

3. Strengthening energy conservation and carbon reduction management in key units

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Clarify the energy conservation and carbon reduction responsibilities of key energy-consuming units and key carbon-emitting units, strictly implement relevant laws and regulations, and establish and improve internal management

Improve management mechanisms, gradually establish energy and carbon emission management systems, and optimize the configuration of energy metering instruments. Strengthen key carbon emission units and key energy users

Construction, improve the professional level of management personnel, carry out energy-saving and low-carbon technology applications and project transformation, strengthen information and intelligent applications, and promote energy consumption

Fine-tuned management. Implement the "leader" action for energy efficiency and carbon emissions, conduct industry benchmarking, and effectively play the exemplary and leading role of advanced units.

III. Better leverage the role of market mechanisms

1. Further improve market mechanisms such as carbon emission rights trading

Give full play to the incentive and constraint role of the carbon market mechanism in reducing carbon emissions, and further optimize the city's pilot carbon market mechanism and supporting measures.

The quota total control system will be relaxed, the coverage of the carbon market will be continuously expanded, the management mechanism of key emission units and the quota allocation method will be improved, and the quota will be enriched.

Further update and improve the carbon market working mechanism, encourage and guide emission units to actively innovate through technology

Management reduces carbon emissions and reduces the overall emission reduction costs of society through market mechanisms. Improve the design of the carbon market system and establish a

Vigorous innovation will be made in energy systems, carbon financial products, and ecological compensation mechanisms to expand the coverage of market mechanisms.

Improve the coordination and unification of working mechanisms and policies such as renewable energy power consumption and energy conservation assessment, carbon emission accounting, and build a mechanism to support and encourage renewable energy.

policy environment.

(II) Make good connections with the national carbon market

In accordance with the overall arrangement of the national carbon market construction, we will do a good job in the management of the city's key emission units that are included in the national carbon market, including those that have been included in the management

Improve the daily supervision mechanism, do a good job in carbon emission reporting, verification, quota issuance and

Process management, strengthen data quality management.

(III) Build and operate a national voluntary emissions trading institution

Under the guidance of the Ministry of Ecology and Environment, we actively promoted the construction of the national greenhouse gas voluntary emission reduction (CCER) trading institution and participated in the formulation of the CCE

Organize the construction of CCER trading and management system, explore the connection and cooperation with international carbon trading and offset mechanisms.

Based on the national voluntary emission reduction trading organization, it will be upgraded to a national green exchange facing the world.

Column 5 Carbon Emissions Trading

Carbon emission rights trading is an important policy measure to control and reduce greenhouse gas emissions through market mechanisms.

Institutional innovation. In layman's terms, it is to treat carbon dioxide emission rights as commodities for trading. Leading technology and high efficiency in energy resource utilization

Since the actual emissions are less than the emission quotas obtained, they can benefit from selling the surplus quotas. On the contrary, high-emission units with backward technology and poor management will benefit from selling the surplus quotas.

If the actual emissions are higher than the quota, it is necessary to purchase quotas or offset products from the market to make up for the shortfall. Through the market mechanism, companies are forced to reduce emissions.

As one of the first seven pilot projects in China, Beijing's carbon emission trading pilot has been running smoothly for eight years since its opening in 2013.

The emission units have reduced emissions through low-carbon energy conversion, new technology utilization and improved management efficiency, which has played a significant role in improving carbon emission performance.

It has played a role in accumulating experience for the construction of the national carbon market.

In July 2021, the national carbon emission trading market was officially launched. More than 2,200 key emission units in the power generation industry became the first to be included in the national carbon

The total carbon emissions of enterprises exceeded 4 billion tons. At the end of 2021, the first compliance period of the national carbon market was successfully completed, with a cumulative transaction volume of 7.661 billion yuan.

As a supplementary mechanism of the carbon market, China Certified Greenhouse Gas Emission Reductions (CCER) can be used as an offset in the national carbon emission rights market.

Key emission units can also directly participate in carbon trading to support relevant institutions in fulfilling their social responsibility of carbon reduction.

The exchange will continue to serve as the trading institution of the pilot carbon market in this city, and also serve as the

It is an important infrastructure for promoting green financial innovation.

4. Promoting energy price reform

Implement the national electricity price reform plan, orderly promote industrial and commercial users to participate in electricity market transactions, and improve the residential tiered electricity price system.

The government will study and establish a heating price management mechanism that is conducive to promoting energy conservation and carbon reduction, green and efficient development. Implement the tiered electricity price policy for high-energy-consuming and high-emission industries.

The national new energy storage pricing mechanism will improve the new energy storage pricing policy and encourage market players to use peak-off-peak electricity price difference, ancillary service compensation and other mechanisms to promote

exhibition.

4. Establish a digital and intelligent management system

1. Strengthening the data foundation

Continue to improve the statistical system for responding to climate change and energy conservation, including sub-sectors and key areas such as construction, heating, transportation and renewable energy.

Integrate statistical resources such as energy and forestry carbon sinks to establish greenhouse gas emission statistics for the city, districts and major industries.

Improve the basic statistical system for greenhouse gas emissions. Form an effective mechanism for sharing energy and greenhouse gas statistical data, and consolidate the energy and greenhouse gas statistics at the municipal and district levels.

Improve the data basis for greenhouse gas emission situation assessment and strengthen the ability to analyze the situation. Improve the accounting, monitoring, reporting and verification of greenhouse gas emission data of emission units.

Focus on strengthening the quality management of carbon emissions data.

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(ii) Compilation of greenhouse gas emission inventories

Establish a regular working mechanism for compiling greenhouse gas inventories with inter-departmental coordination, and compile a city-level greenhouse gas emission inventory that includes various greenhouse gases on an annual basis.

Encourage the preparation of district-level greenhouse gas emissions. Follow the requirements of the relevant inventory compilation methodologies and do a good job in linking the inventory compilation methods and data at all levels.

The accuracy and scientificity of the data in the list will be enhanced, and the list will support the city's decision-making on climate change policies.

3. Establishing a greenhouse gas emission monitoring system

Research and carry out environmental greenhouse gas monitoring and online monitoring of greenhouse gas emissions in key areas, and promote the establishment of greenhouse gas monitoring and ecosystem carbon

system, strengthen the comprehensive analysis and application of monitoring data, and promote the coordinated integration of greenhouse gas statistics and monitoring data.

The innovative model and mechanism of monitoring and verifying greenhouse gas emissions of emission units through big data monitoring and social activities will be used to strengthen the quality of reported data.

Using information technology such as blockchain, we will build a city greenhouse gas emission accounting and management platform, and study and establish a systematic, information-based, and intelligent

Climate change management system to create a new data-driven intelligent governance model.

5. Building a common action pattern for all citizens

Actively organize and carry out publicity activities such as Energy Conservation Week, Low Carbon Day, and Environment Day to intensify publicity efforts on climate change response and energy conservation.

We will carry out education and training on theoretical knowledge related to carbon emissions reduction, energy conservation, carbon peak and carbon neutrality, carry out comprehensive and multi-channel publicity and training, and popularize the scientific concept of energy conservation and carbon reduction.

Strengthen energy conservation and carbon reduction training for managers at all levels and the public. Cultivate a green consumption environment and actively promote energy conservation and high efficiency.

and high-efficiency lighting products, encouraging the public to purchase energy-saving products, cultivate green consumption habits, and practice a low-carbon and environmentally friendly lifestyle.

Mobilize the enthusiasm of the public to participate in green and low-carbon practices. Coordinate the development of energy-saving offices, climate-friendly regions, low-carbon travel, low-carbon buildings and other important projects.

Strengthen media publicity and public supervision, give play to the positive guiding role of the media, and make green living a broad social consensus and conscious action.

Chapter VII Promoting scientific and technological innovation and pilot demonstration

Make full use of the advantages of building an international science and technology innovation center and implement the innovation-driven development strategy in depth.

We will take the improvement of basic capacity as an important part of deepening the work of addressing climate change. We will tap into the source of internal innovation power for green and low-carbon development and optimize green and low-carbon development.

environment, cultivate high-end professional talents, strengthen the pilot demonstration role, and provide support for achieving green and low-carbon development.

1. Accelerate scientific and technological innovation in energy conservation and carbon reduction

1. Increase support for energy-saving and carbon-reduction technology innovation

Strengthen the driving effect of technological innovation on energy conservation and carbon reduction, take the support of carbon peak and carbon neutrality as the guide, and increase the research and development of key technologies.

We will organize and implement a number of major scientific research projects and achieve technological breakthroughs in key areas. Optimize the coordinated dispatching control of source-grid-load-storage

Advances and large-scale applications will promote the integration of power systems and information technology. Accelerate the development of hydrogen energy, high-efficiency photovoltaic and solar thermal, distributed energy storage, and smart grids.

Integrate technologies such as storage and utilization, support innovation in intelligent and diversified energy supply and management systems, and strengthen research on safety technologies for energy storage facilities.

(II) Demonstrate and promote the application of innovative technologies

Accelerate the transformation of energy-saving and carbon-reducing scientific and technological achievements into real productivity. Give full play to the role of the "three cities and one district" as the main platform, and actively promote the certification of energy-saving and carbon-reducing products.

Promote building energy efficiency labeling and certification, and build a wide range of application scenarios in key areas such as low-carbon energy, low-carbon buildings, low-carbon transportation, and heating.

Clean energy system, widely promote energy-saving and carbon-reducing technologies such as industrial heat recovery, heat pump heating, and cascade utilization of park energy.

In the field of energy conservation and carbon reduction, we will vigorously promote the application of innovative technologies in design, materials, equipment technology, and management technology.

We will build a number of professional incubators for energy-saving and carbon-reduction industries in demonstration areas, university science and technology parks, etc. We will track the forefront of technological innovation, establish a dynamic adjustment mechanism, and release

Technology recommendation catalog. Improve the technology and demand docking mechanism, and promote supply and demand docking through environmental protection exhibitions, trade in services and other means. Make full use of the market

We will accelerate the promotion and application of technology and realize the radiation of green industry technology, product export and service output in the capital, and make Beijing a

Carbon peak and carbon neutrality technology innovation base.

3. Strengthening research on major basic issues and professional talent development

Deepen the research on theoretical basis and methods, and actively carry out research on climate change mitigation and adaptation pathways, climate change impact assessment and extreme climate risks.

Basic scientific research on climate change economics, etc. Research and formulate medium- and long-term plans for adapting to climate change. Focus on the technical path and

We will conduct a series of studies on major issues such as the energy system, low-carbon revolution, and synergistic efficiency of pollution reduction and carbon reduction. We will strengthen talent training and establish a development team for energy conservation and carbon reduction.

Encourage colleges and universities to set up disciplines related to responding to climate change, carbon peak and carbon neutrality, and strengthen the cultivation of innovative, application-oriented and skilled talents.

Management personnel and professional talent teams will be established to promote the establishment of district-level support organizations for responding to climate change and energy conservation.

2. Optimize energy-saving and carbon-reduction service industries

1. Actively foster a policy environment conducive to industrial development

Taking Beijing's inclusion in the national business environment innovation pilot program as an opportunity, we will deepen market-oriented reforms, strengthen institutional innovation, and establish an industry that is conducive to energy conservation and carbon reduction.

Policy environment, support to improve the efficiency of energy-saving and carbon-reduction industry. Increase the support of fiscal and taxation policies, and increase the government-led support for enterprises engaged in energy-saving and carbon-reduction technologies.

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Further improve the energy-saving technology reform and contract energy management support policies, and promote

Third-party service mechanisms such as carbon emission management and so on.

(II) Vigorously develop financial services for energy conservation and carbon reduction

Accelerate the cultivation of green financial organizational systems and build digital green financial infrastructure. Encourage financial institutions to introduce energy conservation and carbon reduction evaluation elements.

Green bonds for green industry development, and financial services such as climate-friendly investment and financing. Create green energy-saving and carbon-reduction industries through market-based fundraising.

Encourage and support social funds to invest in the research and development of low-carbon technologies and promote their application. Standardize the development of energy conservation assessment, energy audit, carbon emission evaluation, carbon emission verification, etc.

Improve energy conservation review and certification standards and corresponding platform construction, support banking and financial institutions to carry out equity-bond linkage business in Zhongguancun,

Green credit and green investment. Encourage insurance institutions to conduct climate risk analysis, develop insurance products such as climate disasters, and play a role in climate risk prevention.

Extreme effect.

3. Strengthen the construction of pilot demonstration projects

1. Give full play to the carbon neutrality demonstration role of the Winter Olympics

Taking the carbon neutrality of the Winter Olympics as a large-scale experiment and publicity opportunity to explore and practice the path of carbon neutrality, the carbon neutrality of the Beijing 2022 Winter Olympics will be used as a large-scale

Taking the Green Winter Olympics as a benchmark, we will continue to promote the sustainable management standards of large-scale events in Beijing.

Build and pass on the Winter Olympics sustainable development legacy. Establish carbon neutrality assessment methods and implementation paths for large-scale events.

(II) Strengthening the construction of low-carbon pilot demonstration projects

Formulate the city's low-carbon pilot work plan for the "14th Five-Year Plan" period, and promote the construction of multi-type and multi-level green low-carbon pilot projects throughout the city.

Promote the application of a number of advanced low-carbon technologies, cultivate a number of low-carbon leaders with leading carbon performance, and build a number of climate-friendly

Districts, communities, campuses, etc. Public institutions, key carbon emission units, leading technology companies and other entities are encouraged to actively participate in various low-carbon pilot projects.

3. Carry out climate investment and financing pilot projects

Guide and promote more funds to be invested in the field of climate change response, and lay a financial foundation for the work of addressing climate change.

Make overall arrangements, mobilize and organize implementation of the districts with strong willingness, excellent basic conditions, leading role and typicality, and carry out investment policy guidance,

Support the pilot projects of climate investment and financing. Support the pilot areas to explore differentiated investment and financing models, organizational forms, service methods and management systems.

Restrain high-carbon investment, innovate incentive and constraint mechanisms and linkage mechanisms for funding arrangements, effectively play a leading role in carbon reduction, and promote green development in pilot areas.

Transformation.

Column 6: Climate Investment and Financing Pilot

Climate change investment and financing (hereinafter referred to as "climate investment and financing") refers to the use of

Mobilizing social funds to flow into investment activities in the field of climate change response is an important part of green finance. Climate investment and financing can help stimulate

Open up the market and further promote the formation of energy structures, industrial structures, production methods and lifestyles that mitigate and adapt to climate change.

In October 2020, the Ministry of Ecology and Environment and five other ministries jointly issued the "Guiding Opinions on Promoting Investment and Financing to Address Climate Change" (Environmental Protection and Environment

ÿ2020ÿNo. 57), which clearly stated that the state will vigorously guide the orderly development of climate investment and financing. In December 2021, the Ministry of Ecology and Environment and nine other ministries and commissions

The Notice on Carrying out Pilot Work on Climate Investment and Financing (Environmental Office Climate [2021] No. 27) was issued to encourage all localities to actively organize and encourage

Prefectural-level regions with good foundations and strong representativeness shall apply for climate investment and financing pilot projects.

The city will take the low-carbon pilot demonstration construction as the basis, follow the spirit of national policies, take the development of green finance as an opportunity, guide diversified investment,

We will leverage the support of financial tools in addressing climate change, effectively promote the construction and widespread practice of climate investment and financing pilot projects, and promote the formation of effective

The policy environment for the development of climate investment and financing will be improved, and a group of climate-friendly market players and a climate investment and financing development model with Beijing characteristics will be cultivated.

Actively explore the city's climate investment and financing pilot demonstration and practice based on its characteristics in fields, technologies, etc., and explore participation in international cooperation on climate investment and financing.

Chapter 8 Strengthening the organization and implementation of planning

1. Strengthen organizational leadership

Give full play to the leadership role of the Municipal Committee of Ecology and Civilization Construction in the city's response to climate change and energy conservation.

The Ecological Environment Bureau and the Municipal Development and Reform Commission are the leading units for addressing climate change and energy conservation, respectively, and are responsible for overall design, organization and coordination, and system planning.

All districts, departments and key units should further strengthen the organization and leadership of climate change response and energy conservation, and clearly define the main responsibilities.

One person is responsible, forming a working pattern of pushing forward step by step.

2. Detailed decomposition and implementation

Improve the planning implementation mechanism, strengthen the binding role of energy and carbon emission control targets, and break down the main planning objectives and key tasks into smaller details.

Districts and relevant departments. Each unit shall deploy and implement key tasks according to its division of responsibilities. The district governments and key enterprises shall complete their annual goals and

Conduct follow-up evaluation of the situation. Carry out systematic mid-term evaluation and summary evaluation of the plan to improve the objectivity and scientific nature of the evaluation work of plan implementation.

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3. Guide diversified investment

Increase government funding, improve the financial support mechanism, strengthen municipal financial support and municipal government fixed asset investment support, and actively strive for

Explore the establishment of a market-based special fund for addressing climate change and provide financial support for energy conservation and carbon reduction in key areas.

Encourage and support carbon reduction products.

IV. Deepening exchanges and cooperation

Continue to deepen international cooperation on climate change, including policy dialogue and practical cooperation with international organizations and international cities, and encourage scientific and technological

We will work together to conduct scientific research and technological development, and jointly explore new ways to address climate change. We will actively serve and participate in the "Belt and Road" and "South-South Cooperation" initiatives.

We will strengthen the transfer of green and low-carbon technologies and service output, and participate in supporting and helping less developed regions to improve their capacity to cope with climate change.

We will establish a long-term mechanism for international cooperation, introduce advanced technologies and management concepts, and share them through branded events such as the Beijing International Clean Air and Climate Action Forum.

The achievements of civilization construction tell the story of Beijing's green development well.

Related interpretation

One chart to understand: Beijing Municipal Bureau of Ecology and Environment Beijing Municipal Development and Reform Commission issued the "Beijing 14th Five-Year Plan for Climate Change and Energy Conservation" announcement of

National and provincial and municipal development and reform commissions Beijing Municipal Government Agencies District governments in Beijing Directly affiliated unit website Management unit website News Media

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